

Unraveling Photon Spin

Helicity, Basis Transformations, and Matrix Representations

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PHY 341: Quantum Mechanics | Prof. Brett Fadem

Muhlenberg College | Companion Presentation to Photon Polarization Lesson Plan

Types of Particles: Massive vs. Massless Spin Systems

Electron: Massive Spin-1/2

- ▶ **Rest Mass:** $m_e \approx 9.11 \times 10^{-31} \text{ kg}$
- ▶ **Rest Frame:** Always exists ($v < c$)
- ▶ **Intrinsic Spin:** $s = 1/2$
- ▶ **States ($2s + 1$):** $m_s \in \{+1/2, -1/2\}$
- ▶ **Hilbert Space:** $\mathbb{C}^2$ (SU(2) spinor)

Photon: Massless Spin-1

- ▶ **Rest Mass:** $m_\gamma = 0$ (travels at c)
- ▶ **Rest Frame:** Does not exist
- ▶ **Intrinsic Spin:** $s = 1$
- ▶ **Helicity:** $\lambda \in \{+1, -1\}$
- ▶ **Longitudinal $m = 0$:** **Forbidden** by gauge symmetry

Key Physical Takeaway

Electromagnetic gauge invariance ($\nabla \cdot \mathbf{E} = 0$) eliminates longitudinal polarization ($m = 0$). Propagating photons have **only two physical polarization states** ($|R\rangle$ and $|L\rangle$), forming a 2-level quantum system.

Spatial Rotation Operator and Angular Momentum

Rotations in the Transverse Plane:

- ▶ Rotate electric field vectors by angle θ around propagation axis $\hat{k} = \hat{z}$:

$$R(\theta\hat{k})|x\rangle = \cos\theta|x\rangle + \sin\theta|y\rangle$$

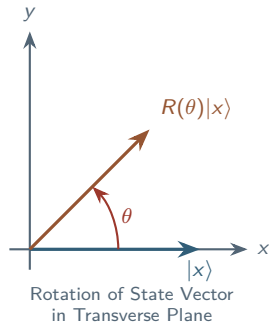
$$R(\theta\hat{k})|y\rangle = -\sin\theta|x\rangle + \cos\theta|y\rangle$$

- ▶ In standard matrix form:

$$R(\theta\hat{k}) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

- ▶ From quantum mechanics, the generator of rotations about the z-axis is the angular momentum operator $\hat{J}_z$:

$$R(\theta\hat{k}) = \exp\left(-\frac{i}{\hbar}\hat{J}_z\theta\right)$$



Two Useful Orthonormal Bases for Photon States

1. Linear Polarization Basis

Basis States: $\{|x\rangle, |y\rangle\}$

$$|x\rangle \longleftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |y\rangle \longleftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- ▶ Orthogonal Cartesian polarization vectors.
- ▶ Convenient for linear polarizers and beam-splitters.
- ▶ **Not** eigenstates of angular momentum $\hat{J}_z$.

2. Circular Polarization Basis

Basis States: $\{|R\rangle, |L\rangle\}$

$$|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle) \implies \hat{J}_z|R\rangle = +\hbar|R\rangle$$

$$|L\rangle = \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle) \implies \hat{J}_z|L\rangle = -\hbar|L\rangle$$

- ▶ Rotating electric field vectors (quarter-cycle phase shift $\pm i$).
- ▶ **True eigenstates** of angular momentum operator $\hat{J}_z$!



$|R\rangle$: **Right-Circular** ($\lambda = +1$)



$|L\rangle$: **Left-Circular** ($\lambda = -1$)

The Quarter-Wave Plate: Physical Preparation of Eigenstates

How Birefringence Prepares $|R\rangle$ and $|L\rangle$:

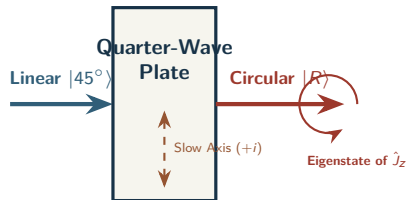
- ▶ Birefringent crystal with orthogonal **Fast** and **Slow** optical axes.
- ▶ Crystal thickness produces a relative phase shift of $\Delta\phi = 90^\circ = \pi/2$ ($e^{i\pi/2} = i$).
- ▶ Input: Linear light at 45° to the axes:

$$|45^\circ\rangle = \frac{1}{\sqrt{2}}(|x\rangle + |y\rangle)$$

- ▶ Setting the vertical axis as the slow axis introduces phase $+i$:

$$|45^\circ\rangle \xrightarrow{\text{QWP}} \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle) = |R\rangle$$

- ▶ If input is at $-45^\circ \implies$ output is $|L\rangle$.



Physical Significance

A macroscopic optical element (QWP) acts directly as a quantum state transformer, converting superpositions into angular momentum eigenstates.

Matrix Representations of the Operator $\hat{J}_z$

Circular Basis: Diagonal

In the basis $\{|R\rangle, |L\rangle\}$:

$$[\hat{J}_z]_{\text{circ}} = \begin{pmatrix} \langle R|\hat{J}_z|R\rangle & \langle R|\hat{J}_z|L\rangle \\ \langle L|\hat{J}_z|R\rangle & \langle L|\hat{J}_z|L\rangle \end{pmatrix}$$

Because $|R\rangle, |L\rangle$ are eigenstates:

$$[\hat{J}_z]_{\text{circ}} = \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix} = \hbar \hat{\sigma}_z$$

Purely diagonal: Eigenvalues sit directly on the diagonal.

Linear Basis: Off-Diagonal

In the basis $\{|x\rangle, |y\rangle\}$:

$$[\hat{J}_z]_{\text{lin}} = \begin{pmatrix} \langle x|\hat{J}_z|x\rangle & \langle x|\hat{J}_z|y\rangle \\ \langle y|\hat{J}_z|x\rangle & \langle y|\hat{J}_z|y\rangle \end{pmatrix}$$

As we will derive:

$$[\hat{J}_z]_{\text{lin}} = \begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix} = \hbar \hat{\sigma}_y$$

Purely off-diagonal: Confirms $|x\rangle, |y\rangle$ are **not** angular momentum eigenstates.

$$\text{Unitary Transformation: } U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \quad [\hat{J}_z]_{\text{lin}} = U[\hat{J}_z]_{\text{circ}}U^\dagger$$

Connecting Classical Waves to Quantum Hilbert Space

The Classical–Quantum Correspondence:

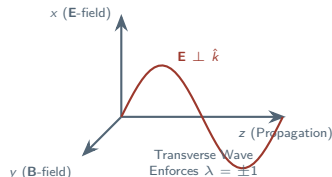
- **Classical View:** Monochromatic light has transverse electric and magnetic fields satisfying Maxwell's equations:

$$\mathbf{E}(z, t) = \text{Re} \left[(E_{0x}\hat{x} + E_{0y}e^{i\delta}\hat{y})e^{i(kz-\omega t)} \right]$$

- **Quantum View:** Each individual photon occupies a state in a 2D Hilbert space:

$$|\psi\rangle = c_x|x\rangle + c_y|y\rangle = c_R|R\rangle + c_L|L\rangle$$

- **Intensity $\longleftrightarrow$ Born Probability:** Classical filter transmission $\propto |E|^2$ corresponds to detection probability $P = |\langle \text{filter} | \psi \rangle|^2$.



Worked Example: Transforming $\hat{J}_z$ into the Linear Basis

Townsend Example 2.8: Calculate $[\hat{J}_z]_{\text{lin}} = U[\hat{J}_z]_{\text{circ}}U^\dagger$.

1. Unitary matrix and adjoint:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \quad U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$$

2. First matrix product $U[\hat{J}_z]_{\text{circ}}$:

$$U[\hat{J}_z]_{\text{circ}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \hbar & -\hbar \\ i\hbar & i\hbar \end{pmatrix}$$

3. Multiply on the right by $U^\dagger$:

$$\begin{aligned} [\hat{J}_z]_{\text{lin}} &= \frac{1}{2} \begin{pmatrix} \hbar & -\hbar \\ i\hbar & i\hbar \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \hbar - \hbar & -i\hbar - i\hbar \\ i\hbar + i\hbar & \hbar - \hbar \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 0 & -2i\hbar \\ 2i\hbar & 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \end{aligned}$$

Interpreting the Result: Invariance of Quantum Physics

What Changed? Representation

- In Circular Basis:

$$[\hat{J}_z]_{\text{circ}} = \hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (\text{Diagonal})$$

- In Linear Basis:

$$[\hat{J}_z]_{\text{lin}} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (\text{Off-Diagonal})$$

- Matrix elements change because the coordinate framework changed.

What Remained Constant? Physics

- **Eigenvalues are preserved:**

$$\det([\hat{J}_z]_{\text{lin}} - \lambda I) = 0 \implies \lambda = \pm \hbar$$

- **Physical Eigenvectors:**

$$[\hat{J}_z]_{\text{lin}}|R\rangle = +\hbar|R\rangle, \quad [\hat{J}_z]_{\text{lin}}|L\rangle = -\hbar|L\rangle$$

- Measurement probabilities are basis-independent.

Core Pedagogical Principle

Matrices depend on the choice of basis; the physical operator does not.

From Polarization to Modern Quantum Optics

The Natural Qubit

- ▶ 2-level quantum state:

$$|0\rangle \equiv |x\rangle, \quad |1\rangle \equiv |y\rangle$$

- ▶ Superpositions physically created using waveplates and beam-splitters.
- ▶ Quantum Key Distribution (BB84 Protocol).

Polarization Entanglement

- ▶ Spontaneous parametric down-conversion (SPDC) produces photon Bell states:

$$|\psi^-\rangle = \frac{|x\rangle_1|y\rangle_2 - |y\rangle_1|x\rangle_2}{\sqrt{2}}$$

- ▶ Rotational singlet: violates Bell's inequalities, confirming quantum non-locality.

Atomic Transitions

- ▶ Hydrogen $2p \rightarrow 1s$ dipole emission (Townsend Ch. 14.7).
- ▶ Selection rule: $\Delta m = \pm 1$.
- ▶ Emitted photon carries angular momentum $\pm\hbar$ (circularly polarized light).

Applications: Quantum Cryptography ■ Quantum Teleportation ■ Astrophysical Spectroscopy

Summary, Pedagogical Reflections, and References

Instructional Summary:

- ▶ Photons are spin-1 bosons, but zero mass restricts physical states to transverse helicities $\lambda = \pm 1$ ($|R\rangle$ and $|L\rangle$).
- ▶ Physical states and observables are independent of coordinate choice; their matrix representations transform unitarily: $[\hat{A}]_B = U[\hat{A}]_A U^\dagger$.
- ▶ Optical hardware (linear polarizers, QWPs) physically perform projection operators and basis transformations.

Teacher Note: This presentation bridges abstract matrix mechanics with laboratory optics, preparing students for both theoretical quantum mechanics and modern quantum technology.

Curricular References

- ▶ Townsend, J. S. *A Modern Approach to Quantum Mechanics*, 2nd ed.; University Science Books, 2012; Chapters 2, 5, and 14.
- ▶ Griffiths, D. J. *Introduction to Quantum Mechanics*, 3rd ed.; Cambridge Univ. Press, 2018.
- ▶ Course: PHY 341 (Quantum Mechanics), Muhlenberg College.

Thank you! Questions & Discussion Welcome.